VI Gaussian random variables and random Processes

Let Gaussian random variable with mean μ and variance σ^2 be denoted

$$W \sim \mathcal{N}(\mu, \sigma^2)$$

The Probability Density Function (PDF) is then of the form

$$f_W(w) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-(w-\mu)^2/2\sigma^2\right]$$

where

$$\mu = E[W] = \int_{-\infty}^{\infty} w f_W(w)\, dw$$

$$\sigma^2 = E[(W-\mu)^2] = E[W^2] - \mu^2$$

Facts about the integrals

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} a \exp\left[-(x-\mu)^2/2\sigma^2\right] dx = f$$

$f = 1$ if $a = 1$

$f = \mu$ if $a = x$

$f = \sigma^2 + \mu^2$ if $a = x^2$

Some further properties of Gaussian Distributions

1) The sum of several random independant variables is Gaussian

$$Y = \sum_{i=1}^{N} \frac{x_i - \mu_i}{\sigma_i}$$

with mean of 0 and variance 1.

2) A Gaussian variable is determined by is mean an variance

A Gaussian process is determed by its mean

$$m_w(t) = E[W(t)]$$

and autocorrelation

$$r_w(t,s) = E[W(t)\bar{W}(s)]$$

A Gaussian Process that with a constant mean and $r_w(t,s) = r_w(s-t)$, is stationary.

3) Linear operators on Gaussian random variables is Gaussian

4) Maximum likelihood detection or estimation involving Gaussian random variables corresponds to a Euclidean distance metric.

5) Wide sense stationary Gaussian random processes are also strict sense stationary

6) Uncorrelated Gaussian random variables are also independant

7) Gausian condition upon a Gaussian is Gaussian

For a Gaussian random vector W with dimension k, mean μ and covariance matrix R

$$W \sim \mathcal{N}(\mu, R)$$

the pdf is

$$f_W(w) = \frac{1}{(2\pi)^{k/2}|R|^{1/2}} \exp\left[-\frac{1}{2}(w-\mu)^T \cdot R^{-1}(w-\mu)\right]$$

$$\mu = E[W] \quad \text{and} \quad R = E[(W-\mu)(W-\mu)^T]$$
$$= E[WW^T] - \mu\mu^T$$

A) Conditional Gaussian Densities

Suppose

$$X \sim \mathcal{N}(\mu_x, \sigma_x^2)$$
$$Y \sim \mathcal{N}(\mu_y, \sigma_y^2)$$

With correlation coefficient, defined as

$$\rho = \frac{E[W_1 W_2] - \mu_1 \mu_2}{\sigma_1 \sigma_2}$$

We want to estimate a value for X, denoted $\hat{x}$. Without measurement, a good guess is

$$\hat{x} = \mu_x$$

Given a signal measurement $Y = y$, we can form a modified prediction using a conditional pdf:

$$f_{X|Y}(x|y) = f(x|y) = \frac{f(x,y)}{f(y)}$$

Where a joint pdf is defined as

$$f(w_1, w_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\frac{(w_1-\mu_1)^2}{\sigma_1^2} + \frac{(w_2-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(w_1-\mu_1)(w_2-\mu_2)}{\sigma_1\sigma_2}\right\}\right]$$

$f(x|y)$ can be written as

$$f(x|y) = \frac{1}{\sqrt{2\pi(1-\rho^2)}\,\sigma_x} \exp\left[\frac{1}{2\sigma_x^2(1-\rho^2)}\left(x - \left(\mu_x + \frac{\sigma_x}{\sigma_y}\rho(y-\mu_y)\right)\right)^2\right]$$

Which requires completing the square to achieve.

We then recognize that $f(x|y)$ is Gaussian with

$$E[x|y] = \mu_x + \frac{\sigma_x}{\sigma_y}\rho(y-\mu_y)$$

$$var(x|y) = \sigma_x^2(1-\rho^2)$$

A reasonable estimate of $\hat{x}$ when $\rho \neq 0$ is

$$\hat{x} = \mu_x + \frac{\sigma_x}{\sigma_y}\rho(y-\mu_y)$$

Also note

$$\sigma_x^2\sqrt{1-\rho^2} \leq \sigma_x^2$$